INTRODUCTION TO PROBABILITY & STATISTICS

A Mini Textbook for Search and Retrieval Demonstration

LibraLink Demo Learning Series

Team-authored sample material created for software testing and demonstration.

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How to use this mini textbook

Each chapter introduces a concept, provides a formula or worked example where useful, and ends with key points. This content is intentionally concise and is designed to test PDF text extraction, passage chunking, and keyword retrieval in LibraLink.

Chapter 1 — Probability and Events

Probability is a numerical way to describe uncertainty. It assigns a value from 0 to 1 to an event, where 0 represents an impossible event and 1 represents a certain event.

1.1 Sample space and events

The **sample space** is the set of all possible outcomes of an experiment. An **event** is a collection of outcomes from that sample space.

For a fair six-sided die, the sample space is {1, 2, 3, 4, 5, 6}. The event of rolling an even number is {2, 4, 6}.

1.2 Basic probability formula

When all outcomes are equally likely, the probability of event A is the number of outcomes favourable to A divided by the total number of possible outcomes.

> **Worked example**
> For a fair die, three outcomes are even and there are six outcomes in total. Therefore, P(even) = 3/6 = 1/2.

1.3 Complement rule

The complement of an event A means that A does not occur. The complement rule is P(not A) = 1 – P(A).

Probability values are always between 0 and 1. The sum of the probabilities of all outcomes in a complete sample space is 1.

Chapter 2 — Random Variables and Distributions

A random variable assigns a numerical value to each outcome of a random experiment. A probability distribution describes the possible values of the variable and the probabilities associated with those values.

2.1 Discrete and continuous variables

A **discrete random variable** takes countable values, such as the number of heads in three coin tosses. A **continuous random variable** can take values across an interval, such as a measured time or height.

2.2 Common probability distributions

The **binomial distribution** models the number of successes in a fixed number of independent trials when each trial has the same success probability.

The **Poisson distribution** is often used to model counts of events in a fixed interval when events occur at a stable average rate.

The **normal distribution** is a continuous, symmetric, bell-shaped distribution characterized by its mean and standard deviation.

2.3 Probability mass function

For a discrete random variable X, the probability mass function gives P(X = x) for each possible value x. Each probability is non-negative, and the probabilities over all possible values sum to 1.

> **Worked example**
> If X is the number of heads in one fair coin toss, then P(X = 0) = 1/2 and P(X = 1) = 1/2.

Chapter 3 — Mean, Variance and Standard Deviation

The mean describes the centre of a distribution. Variance and standard deviation describe how spread out observations are around the mean.

3.1 Mean

For observations x■, x■, …, x■, the arithmetic mean is the sum of the observations divided by the number of observations.

> **Worked example**
> For values 2, 4, and 6, the mean is (2 + 4 + 6) / 3 = 4.

3.2 Variance

Variance measures the average squared deviation from the mean. For a population with values x■ through x■ and population mean μ, variance is the average of (x■ − μ)².

Squaring deviations prevents positive and negative deviations from cancelling and gives greater weight to larger deviations.

3.3 Standard deviation

Standard deviation is the square root of variance. It is expressed in the same units as the original observations, which can make it easier to interpret than variance.

Chapter 4 — The Normal Distribution

The normal distribution is a continuous probability distribution with a bell-shaped curve. It is symmetric about its mean, so the mean, median, and mode coincide in the ideal normal model.

4.1 Parameters

The mean μ determines the centre of the distribution. The standard deviation σ describes its spread. A larger standard deviation produces a wider curve, while a smaller standard deviation produces a narrower

curve.

4.2 Standard scores

A z-score describes how many standard deviations an observation lies above or below the mean. The formula is $z = (x - \mu) / \sigma$, where x is the observation, μ is the mean, and σ is the standard deviation.

> **Worked example**
> Suppose a distribution has mean 50 and standard deviation 10. An observation of 70 has $z = (70 - 50) / 10 = 2$, meaning it is two standard deviations above the mean.

4.3 Interpretation

The total area under a probability density curve is 1. Probabilities for intervals correspond to areas under the curve over those intervals. Exact probabilities and quantiles are typically obtained from a z-table or statistical software.

Chapter 5 — Correlation and Interpretation

Correlation describes the direction and strength of a relationship between two variables. It is a measure of association, not proof that one variable causes the other.

5.1 Correlation coefficient

The Pearson correlation coefficient r ranges from -1 to $+1$. A positive value indicates a positive linear association; a negative value indicates a negative linear association; and a value near zero indicates little linear association.

5.2 Interpreting correlation carefully

Correlation can be affected by outliers and does not necessarily capture nonlinear relationships. A strong correlation alone does not establish a causal relationship. Other variables may influence both observed variables.

5.3 Example

> **Worked example**
> If study time and exam score have a positive correlation in a sample, students who study longer tend to have higher scores in that sample. This observation alone does not prove that study time is the only cause of score differences.

Review Questions

1. What is the difference between a sample space and an event?

2. A fair die is rolled once. What is the probability of obtaining an even number?

3. Explain the difference between a discrete random variable and a continuous random variable.

4. What does variance measure, and how is standard deviation related to variance?

5. A normal distribution has mean 100 and standard deviation 15. Calculate the z-score for x = 130.

6. Why does correlation not necessarily imply causation?

Answer check

2. 1/2. 5. z = (130 – 100) / 15 = 2. The other questions are conceptual and should be answered using the chapter explanations.

End of LibraLink demo mini textbook.